

Sharp Chernoff rates for the nonsmooth family $|\sin x|^\xi$

Candidate full resolution of the periodic numerical question in arXiv:2301.05284

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Status: candidate full result for the source’s periodic family; likely valid; human review needed. For every fixed $t > 0$, the exact uniform-norm decay exponents for both source schemes and every $f_\xi(x) = |\sin x|^\xi$, $\xi > 0$, are determined below.

Source question and precise scope

Katalova (Dragunova), Nikbakht, and Remizov ask what convergence rate remains when the usual domain assumptions for a Chernoff rate theorem fail. They study this numerically for functions of different Hölder smoothness [1, p. 11]. The exact source text is reproduced here.

from theorem 4 that Chernoff functions with higher order of Chernoff tangency to L provide a better rate of convergence. Here we come to the main question that we answer in this paper.

The main question attacked in the paper. What happens if conditions of the above theorem 4 do not hold? For example, what the convergence rate will be if f is differentiable only once but not twice, and f' is e.g. Hölder continuous with Hölder exponent equal to $1/2$? Then $f \notin D(\partial^2)$ and the above theorem say nothing about the rate of convergence. In the present paper, we study this situation numerically for several functions f with different smoothness.

Their principal periodic test family is $f_\xi(x) = |\sin x|^\xi$. The present result gives the sharp asymptotic order for that entire family, for each of their two Chernoff functions. It does not answer their separate variable-coefficient “superfast approximation” open problem.

Let $T(t) = e^{t\partial^2}$ be the heat semigroup on $UC_b(\mathbb{R})$, and set

$$\begin{aligned}(G(s)f)(x) &= \tfrac{1}{2}f(x) + \tfrac{1}{4}f(x + 2\sqrt{s}) + \tfrac{1}{4}f(x - 2\sqrt{s}), \\ (S(s)f)(x) &= \tfrac{2}{3}f(x) + \tfrac{1}{6}f(x + \sqrt{6s}) + \tfrac{1}{6}f(x - \sqrt{6s}).\end{aligned}$$

For $C \in \{G, S\}$ write

$$E_n^C(t, \xi) = \|C(t/n)^n f_\xi - T(t)f_\xi\|_\infty.$$

As usual, $A_n \asymp B_n$ means that their ratio is bounded above and below by positive constants independent of n .

Sharp rate theorem and proof intuition

Theorem 1. *For every fixed $t > 0$ and $\xi > 0$,*

$$E_n^G(t, \xi) \asymp n^{-\min\{1, (\xi+1)/2\}}. \tag{1}$$

For the second-order scheme,

$$E_n^S(t, \xi) \asymp \begin{cases} n^{-2}, & \xi = 2, \\ n^{-\min\{2, (\xi+1)/2\}}, & \xi \neq 2. \end{cases} \quad (2)$$

The constants implicit in $\asymp$ may depend on t and ξ .

Proof intuition. There are two independent obstructions. At every fixed Fourier frequency, the multiplier of $G(t/n)^n$ differs from the heat multiplier by order n^{-1} , while that of $S(t/n)^n$ differs by order n^{-2} . These are the usual tangency ceilings. But the discrete schemes are lattice convolutions, so their multipliers revive to a fixed positive size at frequencies of order $\sqrt{n}$, where the heat multiplier is negligible. The Fourier coefficient of $|\sin x|^\xi$ there has size $(\sqrt{n})^{-\xi-1}$. This gives the second obstruction $n^{-(\xi+1)/2}$. A Gaussian estimate around every revival proves that nothing larger was missed. The lone exceptional case $\xi = 2$ occurs because $\sin^2 x$ is a trigonometric polynomial: it has no high-frequency tail to meet a revival.

Fourier data and a multiplier lemma

Use the absolutely convergent Fourier series

$$f_\xi(x) = \frac{a_0}{2} + \sum_{k \geq 1} a_k \cos(2kx).$$

Beta integration gives

$$a_k = \frac{(-1)^k 2^{1-\xi} \Gamma(\xi+1)}{\Gamma(\xi/2 - k + 1) \Gamma(\xi/2 + k + 1)}. \quad (3)$$

If $\xi \notin 2\mathbb{N}$, reflection and the gamma-ratio asymptotic turn this into

$$a_k = -\frac{2^{1-\xi} \Gamma(\xi+1) \sin(\pi\xi/2)}{\pi} \frac{\Gamma(k - \xi/2)}{\Gamma(k + \xi/2 + 1)}, \quad |a_k| \asymp k^{-\xi-1}. \quad (4)$$

If $\xi = 2m$, the series instead terminates at $k = m$. Notice also from (3) that $a_1 \neq 0$ for every $\xi > 0$.

On the mode $\cos(2kx)$, the three operators have multipliers

$$M_{n,k}^G = \cos^{2n}\left(2k\sqrt{t/n}\right), \quad (5)$$

$$M_{n,k}^S = \left[\frac{2}{3} + \frac{1}{3} \cos\left(2k\sqrt{6t/n}\right)\right]^n, \quad (6)$$

$$H_k = e^{-4tk^2}. \quad (7)$$

Lemma 1 (weighted revival bound). *Let $\beta > 1$. There are constants depending only on t and β such that, for every $n \geq 1$,*

$$\sum_{k \geq 1} k^{-\beta} |M_{n,k}^G - H_k| \leq C(n^{-1} + n^{-\beta/2}), \quad (8)$$

$$\sum_{k \geq 1} k^{-\beta} |M_{n,k}^S - H_k| \leq C(n^{-2} + n^{-\beta/2}). \quad (9)$$

Moreover, if k_n^G and k_n^S are nearest integers to

$$A_G \sqrt{n} = \frac{\pi \sqrt{n}}{2\sqrt{t}}, \quad A_S \sqrt{n} = \frac{\pi \sqrt{n}}{\sqrt{6t}},$$

respectively, then

$$M_{n,k_n^G}^G \geq ct, \quad M_{n,k_n^S}^S \geq ct \quad (10)$$

for all sufficiently large n .

Proof. We give the details because the revival term is exactly what ordinary local Taylor estimates miss. Around the central frequency cell, the analytic expansions of the logarithms, followed by $|e^u - e^v| \leq |u - v|e^{\max(u,v)}$, give

$$|M_{n,k}^G - H_k| \leq Cn^{-1}k^4e^{-ck^2}, \quad (11)$$

$$|M_{n,k}^S - H_k| \leq Cn^{-2}k^6e^{-ck^2} \quad (12)$$

for $k \leq \varepsilon\sqrt{n}$. On the rest of that central cell both sides are exponentially small in n . Thus its contribution to the weighted sums is $O(n^{-1})$ or $O(n^{-2})$.

For the remaining cells use the elementary periodic Gaussian bounds

$$M_{n,k}^G \leq \exp\{-c \operatorname{dist}(k, A_G\sqrt{n}\mathbb{Z})^2\},$$

$$M_{n,k}^S \leq \exp\{-c \operatorname{dist}(k, A_S\sqrt{n}\mathbb{Z})^2\}.$$

They follow from $|\cos u| \leq e^{-c \operatorname{dist}(u, \pi\mathbb{Z})^2}$ and $\frac{2}{3} + \frac{1}{3} \cos u \leq e^{-c \operatorname{dist}(u, 2\pi\mathbb{Z})^2}$. In the cell centered at $\ell A_C\sqrt{n}$, the sum of the Gaussian factor over integer k is bounded uniformly, while $k^{-\beta} \leq C(\ell\sqrt{n})^{-\beta}$. Hence all noncentral cells together contribute at most

$$Cn^{-\beta/2} \sum_{\ell \geq 1} \ell^{-\beta} = O(n^{-\beta/2}).$$

The heat tail is exponentially smaller. This proves (8)–(9).

Finally, rounding the first positive revival center changes the phase by at most $\sqrt{t/n}$ for G and $\sqrt{6t/n}$ for S . Raising the resulting $1 - O(n^{-1})$ quantity to the n th power leaves a positive lower bound, which is (10). $\square$

Proof of Theorem 1

Let $D_n^C = C(t/n)^n f_\xi - T(t)f_\xi$. Its k th cosine coefficient is

$$d_{n,k}^C = a_k(M_{n,k}^C - H_k). \quad (13)$$

If $\xi \notin 2\mathbb{N}$, equations (4) and Lemma 1, with $\beta = \xi + 1$, imply

$$\|D_n^G\|_\infty \leq C(n^{-1} + n^{-(\xi+1)/2}), \quad (14)$$

$$\|D_n^S\|_\infty \leq C(n^{-2} + n^{-(\xi+1)/2}). \quad (15)$$

Both terms are necessary. First, the fixed mode $k = 1$ has

$$M_{n,1}^G - H_1 = -\frac{8t^2}{3}e^{-4t}n^{-1} + O(n^{-2}), \quad (16)$$

$$M_{n,1}^S - H_1 = \frac{64t^3}{15}e^{-4t}n^{-2} + O(n^{-3}). \quad (17)$$

Because $a_1 \neq 0$, these coefficients force lower bounds cn^{-1} and cn^{-2} . Second, at the respective k_n^C from Lemma 1, the heat multiplier is exponentially small, whereas $M_{n,k_n^C}^C \geq c_t$ and $|a_{k_n^C}| \geq cn^{-(\xi+1)/2}$. Thus this single coefficient forces the lower bound $cn^{-(\xi+1)/2}$ for either scheme. Here we used only the standard inequality

$$|d_{n,k}^C| = \left| \frac{2}{\pi} \int_0^\pi D_n^C(x) \cos(2kx) dx \right| \leq 2 \|D_n^C\|_\infty.$$

Taking the larger of the two lower bounds and comparing with (14)–(15) proves the claims for non-even ξ .

If $\xi = 2m$, only finitely many Fourier modes occur. The central-cell estimates and (16)–(17) then give $E_n^G \asymp n^{-1}$ and $E_n^S \asymp n^{-2}$. This agrees with the displayed formulas except that $\xi = 2$ would otherwise incorrectly predict $n^{-3/2}$ for S . The theorem follows. $\square$

What this explains in the source experiments

For $0 < \xi < 1$, both schemes have the same asymptotic exponent $(\xi + 1)/2$: the exact line has slope $1/2$ and intercept $1/2$ as a function of ξ . Once $\xi \geq 1$, G saturates at its formal order 1; S keeps improving until $\xi = 3$ and then saturates at order 2. Consequently, regressions on $4 \leq n \leq 11$ that report an order greater than 1 for G or greater than 2 for S are necessarily pre-asymptotic.

The multiplier revivals also supply a mechanism for the source's observed waves: their centers $A_C\sqrt{n}$ move past the integer Fourier lattice as n changes. The theorem controls the resulting envelope but does not claim that the normalized error has a limit. Exponential initial conditions and the distinct superfast-approximation problem remain outside the result.

Novelty note

The run's four cheap indexes contained no prior result for arXiv:2301.05284. Targeted arXiv searches for the exact heat schemes, the family $|\sin x|^\xi$, and sharp Fourier-multiplier rates found the source and adjacent abstract rate theory, but no primary-source theorem matching Theorem 1. The source's December 2025 revision still lists the phenomena as theoretically unexplained. Novelty is therefore provisional pending expert literature review.

References

- [1] K. A. Katalova (Dragunova), N. Nikbakht, and I. D. Remizov, *Concrete examples of the rate of convergence of Chernoff approximations: numerical results for the heat semigroup and open questions on them*, arXiv:2301.05284 (version downloaded 12 August 2026).